INCH-POUND

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MILITARY SPECIFICATION

# PORK WITH RICE IN BARBECUE SAUCE - SMOKE FLAVORING ADDED, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Department and Agencies for the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers thermostabilized pork with rice in barbecue sauce, in flexible pouches for use as a component of the Meal, Ready-to-Eat (MRE), Individual.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATION

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, SW, Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations (9 CFR Parts 300-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Institutional Meat Purchase Specifications for Fresh Pork, Series 400

(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Room 2603, South Building, Washington, DC 20090-6456.)

U.S. Standards for Grades of Tomato Paste (7 CFR Part 52)

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Room 0709, South Building, Washington, DC 20090-6456.)



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U.S. Standards for Milled Rice (7 CFR Part 68)

(Copies are available from the Federal Grain Inspection Service, APHIS, Printing and Distribution Section, 6505 Belcrest Road, Room G-110, Federal Building, Hyattsville, MD 20782.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.)



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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Pork. Pork shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPS) cuts: Item No. 402B (fresh ham, boneless), Item No. 402C (fresh ham, boneless, trimmed), Item No. 405A (shoulder, picnic, boneless), Item No. 405B (shoulder, picnic, cushion, boneless), Item No. 406A (shoulder, Boston butt, boneless), Item No. 407 (shoulder butt, cellar trimmed, boneless), and Item No. 413 (loin, boneless). All pork shall be certified by a USDA Agricultural Marketing Service (AMS) Agent for condition of the product and compliance with the IMPS quality assurance provisions.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed pork, prior to processing into the finished product, shall be in accordance with the following requirements:

a. Pork processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F, inclusive.

b. Holding in the fresh-chilled state for not more than four days after initial certification is permitted, provided that the pork is maintained in the temperature range of 28°F to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the pork is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence.



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d. Tempering/thawing of frozen pork shall comply with the following:

- microwave tempering/thawing of frozen pork is permitted provided excellent condition of the pork, as determined by the USDA, is maintained
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the pork from the freezer to further preparation
- regardless of the process used, frozen pork shall be tempered/thawed to a temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- pork which has been tempered/thawed shall not be refrozen.

3.2.2 Rice, long-grain, parboiled. Rice shall be parboiled, long-grain milled rice, U.S. No. 2 or better of the U.S. Standards for Milled Rice. The rice shall contain not more than 2 percent of kernels having white ungelatinized areas and not less than 10 percent nor more than 15 percent moisture.

3.2.3 Tomato paste. Tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste.

3.2.4 Water. Water used for ice making, formulation, blanching, washing, and processing shall conform to the National Primary Drinking Water Regulations.

3.2.5 Sugar, brown, light. Brown sugar shall be partially refined cane or beet sugar. The sugar shall be light brown in color and shall possess a sweet, molasses-like flavor.

3.2.6 Vinegar, cider. Cider vinegar shall be prepared from apple cider and be clear, light amber in color, free from haze, sediment, suspended matter and floating particles. The cider vinegar shall contain not less than 5.0 grams of acetic acid per 100 mL at 20°C.

3.2.7 Starch, waxy maize, modified. The starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modifed starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH under shear stress and conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste.

3.2.8 Salt. Salt shall be noniodized, white, refined, sodium chloride, with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.



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3.2.9 Sauce, Worcestershire. Worcestershire sauce shall be a brown to dark brown liquid and shall possess a pleasant, tart, peppery, fruit-spice flavor with a typical heavy viscosity. Titratable acidity (as acetic acid) of the sauce shall be not less than 2.8 percent nor more than 3.3 percent.

3.2.10 Onions, dehydrated, chopped. Dehydrated chopped onions shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.11 Flavoring, pork. Pork flavoring shall be a natural flavor enhancer providing typical oven roasted flavor and aroma notes. It shall be a very dark brown to blackish-brown, opaque liquid having a specific gravity at 25°C of 1.199 to 1.239, a refractive index at 20°C of 1.436 to 1.456, and a flash point of 170°F (see 6.3.1).

3.2.12 Flavoring, smoke. Smoke flavoring shall be an aqueous solution of natural maple and/or oak wood smoke and shall be dark brown-black in color with a clean smoke odor. The smoke flavoring shall have a titratable acidity of 6.2 to 7.2 percent (expressed as acetic acid) a carbonyl level (butanone-2) of 7.0 to 8.0 grams per 100 mL and the smoke flavoring compounds (phenols) shall be 7.5 to 9.0 mg/mL (see 6.3.2).

3.2.13 Mustard flour. Mustard flour shall be a bright yellow powder prepared from a blend derived from the endosperm of the seed of Brassica hirta and Brassica juncea. Mustard flour shall contain not less than 0.4 mL volatile oil per 100 grams of mustard flour and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.14 Garlic powder. Garlic powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.15 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.16 Pepper, red, ground. Ground red pepper shall be derived from red, ripe fruit of Capsicum frutescens L. and shall possess the characteristic yellowish-red to red color. The Scoville Pungency Value shall be not less than 30,000 units. The red pepper shall be uniformly ground to allow a minimum of 95 percent, by weight, to pass through a U.S. Standard No. 40 sieve, and not less than 95 percent, by weight, to be retained on a U.S. Standard No. 60 sieve.

3.2.17 Clove, ground. Ground cloves shall be prepared from the dried, unopened flower buds of Caryophyllus aromaticus L. The powder shall be dark reddish-brown in color and shall possess a strong aromatic odor with a hot pungent taste. The ground cloves shall contain not less than 15.0 mL of volatile oil per 100 grams of ground clove and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.



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3.2.18 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall possess a fragrant, clove-like aroma, strongly aromatic, pungent, clove-like flavor and a dark reddish-brown color. The allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

3.2.19 Cinnamon, ground. Ground cinnamon shall be dried bark of Cinnamomum burmannii Blume commonly known as "Korintji Cinnamon". The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.20 Starch, waxy maize, modified, filling and processing aid. The starch shall be a white to off-white, odorless, finely pulverized powder suitable for use as a filling and processing aid in thermostabilized foods. When hydrated at temperatures of 165°F to 200°F, the starch shall develop a paste. The initial viscosity of the paste shall be retained under conditions of moderate shear and prolonged holding times. During retorting at temperatures in excess of 200°F, the starch shall undergo extensive to complete breakdown of the initial viscosity. The starch shall be bland, shall demonstrate easy dispersibility.

3.2.21 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning mixture shall be labeled with each ingredient and the percentage of each ingredient in the mixture. The ingredients shall be in the same proportions as specified in the barbecue sauce formula (see 3.3.3).

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Pork preparation. Boned and trimmed pork which meets the requirements of 3.2.1 shall be further trimmed, if necessary, to ensure compliance with finished product requirements (see 3.6) and shall be processed as follows:

a. The boned and trimmed pork shall be mechanically ground once through using a grinder plate, having openings measuring 3/4 inch in diameter, and using a four-blade knife. The ground pork shall be uniformly blanched in water until a yield of approximately 75 percent is obtained.

b. Prior to blanching, the ground pork shall be maintained for not more than 12 hours at an internal temperature of 28°F to 40°F.

c. Immediately after blanching, the ground pork shall be incorporated into the product formula (see 3.3.5).



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d. Broth obtained from the blanching operation shall be retained for incorporation into the barbecue sauce (see 3.3.3). The broth shall be maintained in the temperature range of 150°F to 180°F and held for not more than four hours. If the hot broth is not used immediately, the broth shall be cooled to a temperature of 28°F to 40°F within four hours and shall be used within 24 hours.

3.3.2 Rice preparation. The rice shall be prepared as follows:

a. Batch weigh the rice (dry).

b. The rice shall be placed in an excess quantity of 190° to 210°F water and blanched until the rice weighs 1.95 to 2.05 times the original dry weight. NOTE: The blanching must be carefully controlled to ensure compliance with finished product requirements.

c. The blanched rice shall be thoroughly rinsed in an excess of clean cool water to remove all excess rice starch.

d. The cooled, blanched rice shall be thoroughly drained.

e. The blanched rice shall be filled into pouches within two hours after being thoroughly drained.

3.3.3 Preparation of barbecue sauce. The barbecue sauce shall be formulated and prepared as follows:

| Ingredient | Percent by weight |
|---|---|
| Tomato paste (24% natural tomato soluble solids) 1/ | 36.00 |
| Water and broth 2/ | 35.30 |
| Sugar, brown, light | 11.00 |
| Vinegar, cider | 8.30 |
| Starch, waxy maize, modified | 3.00 |
| Salt 3/ | 1.50 |
| Sauce, Worcestershire | 1.50 |
| Onions, dehydrated, chopped | 0.95 |
| Flavoring, pork | 0.50 |
| Flavoring, smoke | 0.50 |
| Mustard flour | 0.50 |
| Garlic powder | 0.50 |
| Lecithin | 0.20 |
| Pepper, red, ground | 0.10 |
| Clove, ground | 0.10 |
| Allspice, ground | 0.03 |
| Cinnamon, ground | 0.02 |

1/ If the specified tomato solids are not available, the tomato solids which are used must be calculated to bring the solids level to that specified and the water shall be adjusted accordingly.



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2/ The pork broth retained during the blanching operation (see 3.3.1) shall be incorporated into the sauce.

3/ The total amount of salt in the sauce formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. A thin slurry shall be made with the starch and part of the water and broth.

b. An emulsion shall be prepared by mixing the lecithin with the water and broth. The emulsion shall be heated with continuous agitation to attain maximum emulsification of the fat in the broth.

c. The starch slurry shall be added to the emulsion and the final mixture heated to 180°F to 190°F with continuous stirring until the mixture has thickened.

d. The balance of the ingredients shall be added. The mixture shall be stirred to ensure the salt is dissolved and the flavorings and spices are uniformly dispersed (caution should be taken to prevent kettle burn-on).

e. The volume of the sauce shall be adjusted with water to compensate for evaporation loss during heating and holding.

f. The sauce shall be filled within four hours after cooking. If a hot fill method is used, the prepared sauce shall be held in the temperature range of 150°F to 180°F at all times prior to filling into the pouches. If a cold fill method is used, the prepared sauce shall be cooled immediately and held at an internal temperature of 28°F to 40°F at all times prior to filling into the pouches.

NOTE: Adherence to the above mixing and heating procedures has been shown to prevent kettle burn-on of the brown sugar and solids in the sauce.

3.3.4 Filling and processing aid starch solution preparation. When a filling and processing aid starch is used, a solution (slurry) shall be prepared with the filling and processing aid starch and part of the sauce formulation water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of product.



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3.3.5 Product preparation. Product preparation to pouch filling and sealing (see 3.4) shall be a continuous process. The pork with rice in barbecue sauce shall be formulated and prepared as follows:

| Ingredient | Percent by weight |
|---|---|
| Pork, ground, blanched | 51.56 |
| Barbecue sauce | 42.19 |
| Rice, blanched | 6.25 |

a. The blanched, ground pork shall be added to the barbecue sauce and mixed only to the extent necessary to produce a uniform mixture of ingredients.

b. The blanched rice shall be added to the pork and barbecue sauce using minimum mixing to form a uniform mixture of ingredients.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 1 of MIL-P-44073.

b. A hot fill system is permitted if compatible with the single-stage or multistage filling and sealing method used.

c. The ingredients may be filled into the pouch using either a single-stage or multistage filling operation.

d. Each filled and sealed pouch shall be in the retort process within two hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value ($F_o$) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. No individual pouch shall have a net weight of less than 7.5 ounces.



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e. The average net weight shall be not less than 8.0 ounces.

f. Pork particles shall be uniformly dispersed throughout the product.

g. Texture of the pork shall not be dry, tough, rubbery, or mushy.

h. Texture of the rice shall not be rubbery, mushy, dry, or hard.

i. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

j. Total weight of blood vessels, cartilage, opaque connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch shall be not more than 0.35 ounces.

k. The product shall show no evidence of excessive heating (materially darkened or scorched).

l. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

m. The drained weight of the pork and rice in any individual pouch shall be not less than 4.0 ounces.

n. The average drained weight of the pork and rice shall be not less than 4.5 ounces.

o. The average fat content of the finished product shall not be greater than 12.0 percent and no individual pouch shall have a fat content greater than 14.0 percent.

p. The salt content of the finished product in any individual pouch shall be not less than 1.0 percent nor greater than 1.5 percent.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The pork component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.



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4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Examination of the filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 7.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 8.0 ounces.

4.5.5 Product and drained weight examination.

4.5.5.1 Product examination. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature ($65^{\circ}F$ to $75^{\circ}F$) after completion of the thermoprocessing operation. The pouches shall be heated in $180^{\circ}F$ to $190^{\circ}F$ water for 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table I. The contents shall be examined for the defects indicated in table II.

TABLE I. Double sampling plan for product examination 1/

| Lot size (pouches) | Sample size (pouches) | Cumulative sample | Acceptance number | Rejection number |
|---|---|---|---|---|
| 0 to 500,000 | 20 | -- | 0 | 2 |
| | 20 | 40 | 1 | 2 |

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.



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c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be examined. Defects found in the first and second sample shall be combined, and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

| Category | Defect |
|---|---|
| Major | |
| 101 | Pork particles not uniformly dispersed throughout the product |
| 102 | Texture of the pork is dry, tough, rubbery, or mushy |
| 103 | Texture of the rice is rubbery, mushy, dry, or hard |
| 104 | Sauce consistency is lumpy, not smooth, and is excessively thick or thin |
| 105 | Total weight of blood vessels, cartilage, opaque connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch is more than 0.35 ounces |
| 106 | Product shows evidence of excessive heating (materially darkened or scorched) |
| 107 | Presence of a bone piece measuring 0.3 inch or more in any dimension |

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).



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4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermostabilization operation. The pouches shall be heated in 180°F to 190°F water for approximately 10 minutes, opened and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the double sampling plan specified in table III. The drained weights shall be determined and inspected for the defects listed in table IV.

TABLE III. Double sampling plan for drained weight inspection 1/

| Lot size (pouches) | Sample size (pouches) | Cumulative sample | Acceptance number | Rejection number |
|---|---|---|---|---|
| 0 to 500,000 | 20 | -- | 0 | 2 |
| | 20 | 40 | 1 | 2 |

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined, and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Drained weight defect 1/

| Category | Defect |
|---|---|
| Major | |
| 101 | Drained weight of pork and rice in a pouch is less than 4.0 ounces 2/ |

1/ To determine drained weight, the pouch contents shall be poured into a flat-bottom container. A minimum of three times the pouch's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated so as to liquefy rendered fat and to remove the sauce. The contents shall then be poured into a U.S. Standard No. 7 sieve in a manner that will distribute



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the product over the sieve. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

2/ If the sample average drained weight of the pork and rice is less than 4.5 ounces, the lot shall be rejected.

4.5.6 Fat and salt content testing. Eight filled and sealed pouches shall be selected at random from the lot. The product shall be tested in accordance with the Official Methods of Analysis of AOAC for fat content, method 976.21, 985.15, or 960.39, and for salt content, method 935.47, except that preparation of the samples shall be as follows: The unopened pouches shall be gently warmed in a water bath to melt fat adhering to the inside of the pouches. The pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Packaging examination. Examination of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units



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able component level. Also, aircraft type, some different components may be repair to the

to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Pork flavoring. It has been found that Natural Flavor Enhancer, Pork Type No. 035 manufactured by Felton International meets the requirements of 3.2.11 and performed satisfactorily in the development of this product.

6.3.2 Smoke flavoring. It has been found that Red Arrow's "Charsol C-6" manufactured by Red Arrow Products Company, meets the requirements of 3.2.12 and performed satisfactorily in the development of this product. The 0.50 percent by weight of smoke flavoring in the sauce formula is based on the titratable acidity and phenol levels specified in 3.2.12. The MRE Pork with Rice in Barbecue Sauce producer is cautioned to verify these requirements prior to sauce formulation.

6.4 Subject term (key word) listing.

Combat field feeding
Food processing
Operational ration
Pouch food
Shelf stable

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8940-0746)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS



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tions, and tests shall be accomplished as re

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-P-44201B | 1993 August 20 |

**3. DOCUMENT TITLE** PORK WITH RICE IN BARBECUE SAUCE - SMOKE FLAVORING ADDED, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE (Include Area Code) (1) Commercial | (2) AUTOVON/DSN |
|---|---|---|
| U.S. Army Natick RD&E Center | 508-651-4501 | 256-4501 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: SATNC-WTP<br>Natick, MA 01760-5018 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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If this requirement is ... to perform ... my